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Technical Skills

Languages TypeScript, Python	Frontend React, Next.js, Tailwind CSS	Backend Node.js, Express.js, REST APIs, Event-Driven Architecture, Async Processing
Databases PostgreSQL, Redis, MongoDB	Async & Messaging BullMQ, Dead-Letter Queues, Idempotent Processing	Infrastructure AWS, Docker, GitHub Actions CI/CD, Vercel, DigitalOcean

Work Experience

Vogel Edits 01/2025 - Present
Full-Stack Engineer (Adobe Extensions)

- Shipped production Adobe After Effects & Premiere Pro extensions with **React** UI and **Node.js** tooling, including Vanish BG (real-time AI background removal) published on <https://aescrpts.com>.
- Designed a **Node.js** sidebar architecture to bypass Adobe CEP sandbox limits, using local HTTP IPC to stream inference outputs to the UI in real time.
- Implemented local model inference workflows (**ONNX**) and integrated them into user-facing editing pipelines with low-latency feedback loops.
- Collaborated asynchronously with designers and stakeholders, delivering production-ready features with clear ownership and iterative releases.

Freelance 11/2024 - 01/2025
Full-Stack Engineer

- Built a full-stack tourism booking platform using **Next.js**, **TypeScript**, and **Node.js** with type-safe **PostgreSQL** access via TypeORM.
- Improved backend performance using **PostgreSQL** indexing and **EXPLAIN ANALYZE**, achieving sub-100ms API response times under concurrent load.
- Integrated Paymob payments with **idempotent** transaction handling and secure webhook verification, reducing checkout friction by 40%.
- Raised SEO and Core Web Vitals to 95+ Lighthouse via **Next.js** SSR, image optimization, and lazy loading.

Projects

RootCluster — AI-Powered Log Analysis Platform <https://rootcluster.dev>
Tech Stack: Node.js, TypeScript, PostgreSQL (pgvector), Redis, BullMQ, ONNX, AWS ECS Fargate, Docker, GitHub Actions

- Designed and deployed a distributed, **event-driven** ingestion pipeline on **AWS ECS Fargate** using **BullMQ** for job queuing, **Redis** for caching/rate limiting, and **PostgreSQL (pgvector)** for similarity search, processing 1M+ events/hour with sub-second latency.
- Implemented robust **async processing** patterns: producer-consumer queues, worker pools with configurable concurrency, and priority routing for urgent vs. backfill workloads.
- Built **dead-letter queues** with retry policies (exponential backoff, max attempts) to isolate malformed events without blocking the pipeline.
- Designed **idempotent** processing with Redis-backed dedup keys and at-least-once delivery semantics via job completion callbacks.
- Reduced **ONNX** embedding time from 20+ minutes to under 1 minute via batching, caching, and worker autoscaling.
- Delivered CI/CD using **GitHub Actions** with AWS OIDC to ECR, plus RDS PostgreSQL, ElastiCache, SSM Parameter Store, and automated deployments.

Medical CRM — AI-Enhanced Prescription Platform

Tech Stack: React, Node.js, PostgreSQL

- Built a full-stack CRM for diagnoses, prescriptions, and patient records with **React** and **Node.js** backed by

PostgreSQL.

- Implemented secure authentication with **JWT** access/refresh token rotation and **RBAC** across three permission levels.
- Designed drug interaction validation across 500+ medications using graph-based **PostgreSQL** lookups.
- Optimized dashboard queries for 10K+ monthly records using composite indexes and window functions.

Coursework

- Meta Certified Frontend Developer — Meta (2024)
- Complete SQL Databases Bootcamp: Zero to Mastery — Udemy
- Master the Coding Interview: Data Structures & Algorithms — Udemy
- Node.js: The Complete Guide — Udemy
- Grokking Algorithms — Aditya Bhargava
- Designing Data-Intensive Applications — Martin Kleppmann